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UCS.py - C:/Users/yasha/Downloads/UCS.py (3.12.10)
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import heapq

graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('D', 2), ('E', 5)],
    'C': [('F', 1)],
    'D': [],
    'E': [('G', 1)],
    'F': [('G', 3)],
    'G': []
}

def ucs(start, goal):
    pq = [(0, start)]
    visited = set()

    while pq:
        cost, node = heapq.heappop(pq)

        if node == goal:
            return cost

        if node not in visited:
            visited.add(node)

            for neighbour, weight in graph[node]:
                if neighbour not in visited:
                    heapq.heappush(pq, (cost + weight, neighbour))

print("Minimum Cost:", ucs('A', 'G'))
```

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IDLE Shell 3.12.10
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Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/yasha/Downloads/UCS.py =====
>>> Minimum Cost: 7
>>> |
```